

CENK CELIK

cenk.celik@ucl.ac.uk • cenk.celik@proton.me • ORCID • Google Scholar • London WC1E 6BT

Computational cancer biologist applying ecological and systems-level thinking to understand tumour cell plasticity and microenvironment interactions

Current Position

Research Fellow, University College London, UK (Jun 2023–present)

Education

PhD in Medicine, National University of Singapore, Singapore (Aug 2015–Aug 2020)

- Investigated biophysical regulation of stem cell fate, elucidating how electromagnetic cues and scaffold architecture coordinate chondrogenic differentiation

MSc in Bioengineering, Ege University, Türkiye (July 2012–May 2015)

- Produced and characterised hybrid tissue scaffold approaches for 3D vascularised bone tissue

BSc in Bioengineering, Ege University, Türkiye (Sep 2007–Jun 2012)

- Received Republic of Türkiye Prime Minister Scholarship (stipend, 4 years)
- Awarded and administered TUBITAK 2209 University Students Research Projects Support Programme (₺1,000)

Professional Experience

Research Fellow, University College London, UK (Jun 2023–present)

- Developed EnrichMap, a spatially informed enrichment analysis framework for spatial -omics biology
- Leading a research programme investigating G0 cell cycle arrest in breast cancer, integrating single-cell and spatial transcriptomics across multiple cohorts to map the spatial ecology of proliferative and dormant tumour states
- Contributed to multiple projects (EMT and TAMs in breast cancer, LLG, CRC, osteosarcoma) within UCL, King's and the Crick and international teams
- Co-supervised 2 MSc students and mentored 3 PhD students through weekly one-on-one meetings

Research Fellow, Nanyang Technological University, Singapore (Mar 2021–May 2023)

- Led an independent project elucidating host-pathogen synergy in wound healing and infection, identifying and characterising a previously unrecognised anti-inflammatory macrophage state associated with persistent infection
- Conducted and analysed NGS data by developing pipelines for model organisms and cell line models for different projects focusing on pathologies of ER stress

Research Assistant, National University Hospital System, Singapore (Nov 2019–Aug 2020, short-term)

- Planned, modified and executed research techniques, procedures and tests for nanofibrous scaffold fabrication

Publications (18 publications, 7 first author, *corresponding author, † key papers)

Research articles

- Evans L, Conde-Poole M, **Celik C**, ..., Secrier M, Barral P (2026). CD1d remodels the tumour-infiltrating myeloid populations controlling anti-tumour immunity. *J Exp Med*
- † **Celik C**, ... , Secrier M (2026). Spatial ecology of breast cancer reveals co-evolution of proliferative and dormant niches. *Genome Med* [[10.1186/s13073-026-01711-0](https://doi.org/10.1186/s13073-026-01711-0)]
- da Silva RAG, Tien BYQ, Hsien PNK, **Celik C**, ... , Thibault G, Chen J, Kline KA (2026). *Enterococcus faecalis*-derived lactic acid facilitates persistent and polymicrobial wound infections by suppressing macrophage activation. *Cell Host & Microbe* 34, 1-18. [[10.1016/j.chom.2026.01.002](https://doi.org/10.1016/j.chom.2026.01.002)]
- Tan AMZ, **Celik C**, ... , Thibault G, Kline KA (2026). *Enterococcus faecalis* redox metabolism activates the unfolded protein response to impair wound healing. *Sci Adv*. eab5297. [[10.1126/sciadv.aeb5297](https://doi.org/10.1126/sciadv.aeb5297)]
- Pan S, Withnell E, **Celik C**, Secrier M (2026). From cell types to cell states: adapting foundation models for cellular plasticity. *bioRxiv* [[10.1101/2024.08.16.608311](https://doi.org/10.1101/2024.08.16.608311)]
- Withnell E, **Celik C**, Secrier M (2025). Integrative spatial modelling of cellular plasticity using graph neural networks and geostatistics. *bioRxiv* [[10.1101/2025.09.24.678189](https://doi.org/10.1101/2025.09.24.678189)]
- † **Celik C*** & Secrier M (2025). EnrichMap: Spatially informed enrichment analysis for functional interpretation of spatial transcriptomics. [[10.1101/2025.05.30.656960](https://doi.org/10.1101/2025.05.30.656960)] *In revision*
- † **Celik C**, ... , Kline KA, Thibault G (2024). Decoding complexity of delayed wound healing following *Enterococcus faecalis* infection. *eLife* 13:RP95113. [[10.7554/eLife.95113.3](https://doi.org/10.7554/eLife.95113.3); Selected for [eLife Digest](#)]
- Kaliya-Perumal AK, **Celik C**, ... , Ingham PW (2024). Genetic regulation of injury induced heterotopic ossification in adult zebrafish. *Dis Model Mech* dmm.050724. [[10.1242/dmm.050724](https://doi.org/10.1242/dmm.050724)]

- Dudkevich R, Koh JH, Beaudoin-Chabot C, **Celik C**, ... , Thibault G, Henis-Korenblit S (2022). Neuronal IRE-1 coordinates an organism-wide cold-stress response by regulating fat metabolism. *Cell Reports* 41, 111739. [[10.1016/j.celrep.2022.111739](https://doi.org/10.1016/j.celrep.2022.111739)]
- Beaudoin-Chabot C, Wang L, **Celik C**, ... , Thibault G (2022). The unfolded protein response reverses the effects of glucose on lifespan in chemically-sterilised *C. elegans*. *Nat Commun* 13, 5889. [[10.1038/s41467-022-33630-0](https://doi.org/10.1038/s41467-022-33630-0)]
- Yang Z, **Celik C**, Parate D, Franco-Obregón A, Lee EH (2022). The application of pulsed electromagnetic field for cartilage regeneration. *Tissue Eng Part A* 28, S–365. [[10.1089/ten.tea.2022.29025](https://doi.org/10.1089/ten.tea.2022.29025)]
- **Celik C**, ... , Yang Z (2021). Directionalities of magnetic fields and topographic scaffolds synergise to enhance MSC chondrogenesis. *Acta Biomater* 119, 169–183. [[10.1016/j.actbio.2020.10.039](https://doi.org/10.1016/j.actbio.2020.10.039)]
- Parate D, Kadir ND, **Celik C**, ... , Franco-Obregón A & Yang Z (2020). Pulsed electromagnetic fields potentiate the paracrine function of mesenchymal stem cells for cartilage regeneration. *Stem Cell Res & Ther* 11:46. [[10.1186/s13287-020-1566-5](https://doi.org/10.1186/s13287-020-1566-5)]
- Cetmi SD, ... , **Celik C** & Oncel SS (2019). Preparation of electrospun polycaprolactone nanofiber mats loaded with microalgal extracts. *Eng Life Sci* 19:691–699. [[10.1002/elsc.201900009](https://doi.org/10.1002/elsc.201900009)]
- Duman OM, **Celik C**, Sarikanat M & Sendemir A (2014). Investigation of central nervous system neurons under mechanical strain: An *in vitro* traumatic brain injury model. *18th National Biomedical Engineering Meeting*, 1–4. [[10.1109/biyoumut.2014.7026384](https://doi.org/10.1109/biyoumut.2014.7026384)]

Reviews & Book Chapters

- **Celik C**, ... , Secrier M (2025). Decrypting cancer's spatial code: from single cells to tissue niches. *Mol Oncol* 19: 3465–3485. [[10.1002/1878-0261.70100](https://doi.org/10.1002/1878-0261.70100)]
- **Celik C**, ... , Thibault G (2023). Endoplasmic reticulum stress and lipids in health and disease. *Progress in Lipid Research* 8, 101198. [[10.1016/j.plipres.2022.101198](https://doi.org/10.1016/j.plipres.2022.101198)]
- **Celik C**, ... , Toh WS (2018). Injectable Hydrogels for Cartilage Regeneration. In: Thakur V, Thakur M (eds) *Hydrogels. Gels Horizons: From Science to Smart Materials*. Springer, Singapore. [[10.1007/978-981-10-6077-9_12](https://doi.org/10.1007/978-981-10-6077-9_12)]
- Karaman O, **Celik C** & Sendemir A (2016). Self-Assembled Biomimetic Scaffolds for Bone Tissue Engineering. In Management Association, I. (Ed.), *Biomedical Engineering: Concepts, Methodologies, Tools and Applications*, 476–504, Hershey, PA: IGI Global. [[10.4018/978-1-5225-3158-6.ch021](https://doi.org/10.4018/978-1-5225-3158-6.ch021)]

Research Funding

- **Named Researcher**. UKRI Future Leaders Fellowship MR/Y034031/1 (2025–present)
“Charting the ecosystem of dormancy and persistence after therapy in cancer”, PI: Maria Secrier, UCL
- **Singapore International Graduate Award**. S\$200,000, NUS, Singapore (2015–2019, 4 years)
- **Research Scholarship**. TUBITAK 1002 Short-term R&D Funding, Türkiye (2014–2015, 11 months)
- **Research Scholarship**. TUBITAK Career Grant, Türkiye (2012–2013, 1 year)
- **Research Project Support Programme for Undergraduate Students**. ₺1,000, Türkiye (2011–2012, 16 months)

Awards, Honours & Funding

- **Crick Cancer Symposium 2024**, Selected talk, London (Oct 2024)
- **Poster Presentation Winner**, Festival of Genomics & Biodata, London (Jan 2024)
- **Translational Researcher Programme**, 10X Genomics, Inc., Singapore (Feb 2022)
- **Young Orthopaedic Researcher**, APOA Sports Meeting '19, Kuala Lumpur (Apr 2019)
- **Researcher Travel Grant**, National University of Singapore, Singapore (Sep 2018)
- **Prime Minister Scholarship for Undergraduate Students**, Türkiye (2007–2012, 5 years)

Invited talks & lectures

- **Mapping cell state switches in adaptive cancer niche with spatial transcriptomics**. Guy's Cancer Forum, KCL (24 Mar 2026), London, UK
- **The broken circle breakdown**. External seminar, Ege University (11 Mar 2026), İzmir, Türkiye
- **Exploring spatial patterns of cancer evolution**. Guest lecturer, Ege University (2 Mar 2026), İzmir, Türkiye
- **Spatial evolution of proliferative and arrested niches in breast cancer**. *AI in Pharma*, Hacettepe University (10 May 2025), Ankara, Türkiye
- **Internal and external regulation of cell cycle arrest in breast cancer**. Izmir Biomedicine and Genome Centre (23 Sep 2024), İzmir, Türkiye
- **Intrinsic and extrinsic regulation of cell cycle arrest in breast cancer**. Crick Cancer Interest Group (19 Jun 2024), London, UK

Proceedings

- **Celik C**, Weston WA, de Moreas-Lacerda T, Withnell E, Pan S, Chu T, Labbadia J, Barr AR, Secrier M (8–11 Jun 2026). Spatial ecology of proliferative and dormant niches in breast cancer [Poster]. *EACR Congress 2026*, Budapest, Hungary
- **Celik C**, Weston WA, de Moreas-Lacerda T, Withnell E, Pan S, Chu T, Labbadia J, Barr AR, Secrier M (21–22 Apr 2026). Spatial ecology of proliferative and dormant niches in breast cancer [Poster]. *International Society for Computational Biology UK*, Cambridge, UK
- Champney E, **Celik C**, Zaccaria S, Secrier M (21–22 Apr 2026). Spatial and evolutionary mapping reveal dynamics of homologous recombination deficiency in breast cancer [Poster]. *International Society for Computational Biology UK*, Cambridge, UK
- **Celik C**, Withnell E, Secrier M (7 Oct 2024). Intrinsic and extrinsic factors influencing G0 cell cycle arrest in breast cancer [Selected talk]. *Crick Cancer Research Symposium 2024*, London, UK
- Kaliya-Perumal AK, **Celik C**, Carney T, Harris MP, Ingham PW (17–21 Aug 2024). Injury-induced Heterotopic Ossification: Insights from a novel zebrafish model [Poster]. *18th International Zebrafish Conference*, Kyoto, Japan
- **Celik C**, Withnell E, Secrier M (1 May 2024). Unravelling internal and external regulatory mechanisms of cell cycle arrest in breast cancer [Poster]. *UCL Cancer Symposium 2024*, London, UK
- **Celik C**, Secrier M (24–25 Jan 2024). Deciphering the interplay between cell cycle arrest and tumour microenvironment in breast cancer [Selected talk & Poster]. *Festival of Genomics & Biodata*, London, UK
- **Celik C**, Secrier M (8–10 Oct 2023). Deciphering the interplay between cell cycle arrest and tumour microenvironment in breast cancer: Insights from single-cell data [Poster]. *4th Crick International Cancer Conference*, London, UK
- Lee STL, **Celik C**, ... , Kline KA, Thibault G (3–7 Dec 2022). Dissecting the synergistic role of the unfolded protein response in wound infections [Poster & Talk]. *ACSB & EMBO Meeting*, Washington DC, USA
- **Celik C**, ... , Hui JHP (4–7 Apr 2019). Altering MSC chondrogenesis by PEMF [Talk]. *APOA Sports Meeting 19*, Kuala Lumpur, Malaysia
- **Celik C**, ... , Hui JHP (9–11 Sep 2018). Enhancement of MSC chondrogenesis by PEMF [Poster]. *5th TERMIS World Congress*, Kyoto, Japan
- Cetmi SD, ... , **Celik C**, Sendemir A & Oncel SS (11 Nov 2015). Utilisation of microalgal extracts for construction of tissue scaffolds with electrospinning techniques [Poster]. *7th International Bioengineering Congress*, İzmir, Türkiye
- Duman OM, ... , **Celik C** & Sendemir A (11 Nov 2013). Fe₃O₄ reinforced polycaprolactone nanofibrous scaffolds [Poster]. *Vth International Bioengineering Congress: Human Welfare*, Kuşadası, Türkiye
- **Celik C**, Karaman O, Sendemir A (11 Nov 2013). More than a picture: Biomedical Illustration [Talk]. *19th International Biomedical Science and Technology Symposium*, Kuşadası, Türkiye
- **Celik C**, Gorgun C, Sendemir A (11 Nov 2013). Cell viability of keratinocytes on electrospun Spirulina/PCL composites [Poster]. *19th International Biomedical Science and Technology Symposium*, Kuşadası, Türkiye
- Duman OM, ... , **Celik C** & Sendemir A (9 Sep 2013). PCL/Fe₃O₄ Scaffold Production with Electrospinning [Oral]. *Advanced Materials World Congress 2013*, Çesme, Türkiye
- Serdengecti C, ... , **Celik C**, Sendemir A & Seki Y (4 Apr 2013). Production and analysis of parameters affect morphological characters of chitosan nanospheres [Poster]. *1st Ege University Nanotechnology Days*, İzmir, Türkiye
- Duman OM, **Celik C**, Minaz MC, Sendemir A (4 Apr 2013). Fabrication of PEO/Fe₃O₄ composite tissue scaffolds by electrospinning [Poster]. *1st Ege University Nanotechnology Days*, İzmir, Türkiye
- Demirkaya C, **Celik C**, ... , Sendemir A (5 May 2012). Applications of microalgae for tissue engineering [Oral]. *Marine Biotechnology and Genomics Workshop*, Bodrum, Türkiye
- **Celik C**, ... , Sendemir A (11 Nov 2011). Effects of microalgae reinforcement on biocompatibility and mechanical properties of tissue engineering scaffolds [Talk]. *1st National Aegean Composite Materials Symposium*, Selçuk, Türkiye

Professional Service

Peer review

- Nature Communications (2 reviews)

Memberships

International Society for Computational Biology (2026–present); British Association for Cancer Research (2025–present); European Association for Cancer Research (2024–present); The Genetics Society UK (2023–2024); Society for Cell Biology Singapore (2022–2023)

Teaching & Outreach

- **Reimagining Health with Big Data**. Norwich Science Festival (14 Feb 2026), Norwich, UK
- **Life at the single-cell and spatial scale**. Bootcamp, Bilimler Köyü (10–17 Aug 2025), Foça, Türkiye
- **Art of ATGC: Genetic data analysis and visualisation**. Bootcamp, Bilimler Köyü (11–18 Aug 2024), Foça, Türkiye
- **Coding the code**. Multiple virtual bootcamps, KizCode (Girls who code) (Oct 2021–Jan 2024), Online
- **Graduate Teaching Assistantship Programme**. (2016–2018, 3 years), National University of Singapore, Singapore

Organisational Experience

- **Seminar organiser**, UCL Genetics Institute Seminars (Mar 2026–present), London, UK
- **Founding organiser**, UCL Computational Research Exchange Cancer Network (Dec 2024–Mar 2025), London, UK
- **Hackaton assistant**, EMBO Spatial Biology of Cancer Workshop (Sep 2024), London, UK
- **Committee member**, London Postdoc Network (Apr 2024–present), London, UK
- **Executive committee member**, Singapore–Turkey Friendship Association (May 2016–Jun 2018), Singapore
- **Organising committee member**, 19th International Biomedical Science and Technology Symposium & VIth International Bioengineering Congress: Human Welfare (Nov 2013), Kuşadası, Türkiye

Technical Skills

Computational: Development of analytical frameworks for single-cell and spatial transcriptomics; graph-based and spatial modelling; high-dimensional RNA-seq analysis; reproducible workflows in Python and R; pipeline development and automation (Bash); version-controlled environments (Git); visual communication and scientific figure design

Experimental: Design and execution of bulk and single-cell RNA-seq experiments; molecular and cellular assays (qPCR, western blot, immunofluorescence, immunohistochemistry); mammalian cell culture and viral transduction; small animal models; flow cytometry; experimental design and quantitative data analysis

References

- **Maria Secrier**, Associate Professor, University College London, UK, m.secrier@ucl.ac.uk
- **Guillaume Thibault**, Associate Professor, Nanyang Technological University, Singapore, thibault@ntu.edu.sg